



Production and MRP: Process Accounting

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the Production Process Accounting course topic.

Please complete the Production Process course topic before taking this training.

In addition, please note that this course is relevant for perpetual inventory managed companies.

Objectives



At the end of this topic, you will be able to:

- Describe the journal entries created in the production process.
- Explain the changes made in the inventory and variance accounts during production process.
- Explain the effect of resources on the produced items cost.
- Explain the summary information displayed in the *Production Order* and the variance report.

After completing this topic, you will be able to:

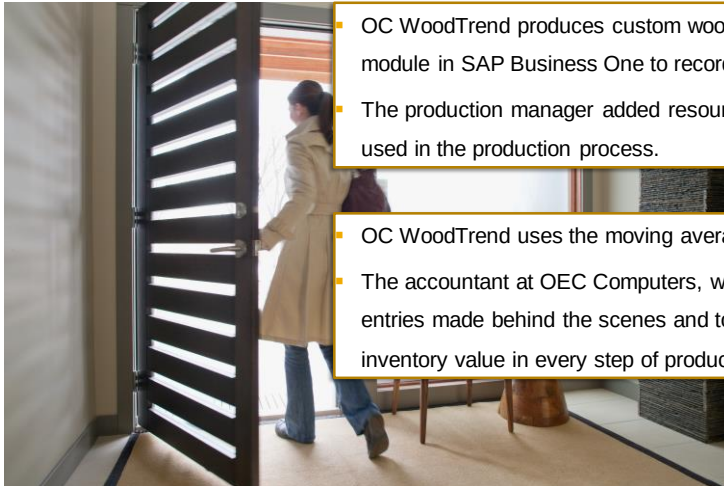
Describe the journal entries created in the production process.

Explain the changes made in the inventory and variance accounts during production process.

Explain the effect of resources on the produced items cost.

Explain the summary information displayed in the Production Order and the variance report.

Business Scenario



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- OC WoodTrend produces custom wooden doors. They use the *Production* module in SAP Business One to record the production process.
- The production manager added resources for the employees and machines used in the production process.
- OC WoodTrend uses the moving average method to manage their inventory.
- The accountant at OEC Computers, would like to learn about the journal entries made behind the scenes and to understand what happens to the inventory value in every step of production.

3

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Production Process Postings – Basic Concept

No Cost Variance

1

Issue for Production Journal Entry

Account	Debit	Credit
Inventory account (Components)		100
WIP Inventory account (Components)	100	
Resource Exp. account		50
WIP Resource account	50	

2

Receipt from Production Journal Entry

Account	Debit	Credit
Inventory account (Produced Item)	150	
WIP Inventory account (Produced Item)		150

3

Production Order Closing Journal Entry

Account	Debit	Credit
WIP Inventory account (Components)		100
WIP Inventory variance account	100	
WIP Inventory account (Produced Item)	150	
WIP Inventory variance account		150
WIP Resource account		50
WIP Inventory variance account	50	

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4

In OC WoodTrend, a Production Order was entered for a decorative wooden door Bill of Materials.

The production manager releases the Production Order and issues the components. The system posts the cost of the components to the debit side of the WIP (Work In Progress) account and to the credit side of the inventory account (1). When involving resources in the production order, a WIP resource account is debited and a resource expense (cost) account is credited.

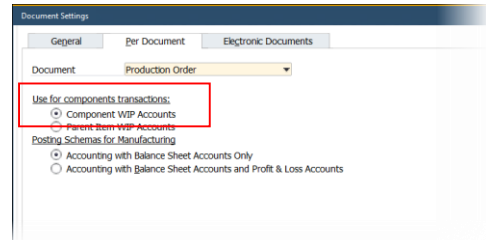
When production is done, the production manager, adds a Receipt from Production document. In the journal entry created, the inventory account is debited with the parent item cost, against the WIP inventory account.

Finally, the Production manager closes the Production Order. This time, the system posts another journal entry. The purpose of this journal entry is to zero down the WIP account balances. In our example, the WIP Inventory account balance, after adding the Issue for Production was 100 in debit. Then, in the Receipt from Production document, this account was credited with 150. The balance of the account before the closing journal entry was 50 in credit and therefore, the account was debited in the amount of 50. The WIP resource account however, had a balance of 50 in debit prior to the closing journal entry. This is why the account was in credit for the amount of 50.

Note that when defining the same accounts for the components and produced item, the amounts may be consolidated into one row.

WIP Accounts

- A WIP inventory account holds the inventory value of the components in the time of production.
- Default accounts in the documents derive from *the Item/ Item Group/ Warehouse Master Data* or defined in a rule in the *Advance GL Account Determination*.
- In the document settings, we define which WIP account to use in the component transactions – components or parent.



A WIP inventory account holds the inventory value of the components in the time of production.

By default, the WIP account and WIP Variance account are determined according to the defaults in the item/ Item group/ Warehouse master data, depending on the inventory method used in the company. When working with the Advanced GL Account Determination, the accounts are determined according to the rules supplied. However, we can manually enter a unique account for each row in the Bill of Materials and in the Production Order.

The WIP-accounts used for components can either be set to use the WIP-accounts for the produced item or the WIP-accounts for the component item, depending on a parameter in the document settings for the production order, as shown in the image.

Production Process Postings – Basic Concept Including Cost Variance

1

Receipt from Production Journal Entry

Account	Debit	Credit
WIP Inventory Account (Produced Item)		150
inventory account (Produced Item)	150	

2

Issue for Production Journal Entry

Account	Debit	Credit
Inventory Account (Components)		120
WIP Inventory Account (Components)	120	
Resource Exp. Account		50
WIP Resource Account	50	

3

Production Order Closing Journal Entry

Account	Debit	Credit
WIP Inventory Account (Produced Item)	150	
WIP Inventory variance account		150
WIP Inventory Account (Components)		120
WIP Inventory variance account	120	
WIP Resource Account		50
WIP Inventory variance account	50	
Inventory account (Produced Item)	20	
WIP Inventory variance account		20

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6

In some cases there may be a variance in the cost value of the components and the parent item during the production process. This can happen when working with the *Manual* issue method. An example scenario for that can be when the parent item is received prior to the components items issue and the cost of the components have changed before they are issued from inventory.

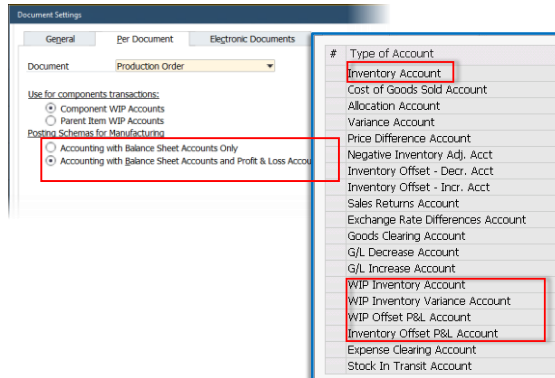
When variance occurs, the system adjusts the inventory account value for the produced item for the variance amount in the closing journal entry, if the produced item is still in inventory.

In the image we can see that the cost of the produced item, as shown in the *Receipt from Production* journal entry is **150**. However, the total cost of the component items and resource is **170**.

When this variance occurs, the system adds another two rows in the closing journal entry to balance of the Inventory account against the WIP variance account.

Note that the variance accounts refer to the produced item

WIP and Inventory Offset P&L Accounts



- P&L offset accounts can be used in production postings
- This account is the offset accounts to the Inventory and WIP accounts
- Suitable for legal requirement in some EU countries (CZ, SK, HU)
- Business practice in some organizations
- Key Benefit: Enhanced monitoring possibilities

The WIP accounts we got to know in the former slides are balance sheet accounts, however we can involve P&L (Profit and Loss) accounts as well if required.

In addition to the traditional journal entries we saw, two more intermediate journal entries are created involving a WIP offset P&L account and an inventory offset P&L account.

This solution is required in several countries and allows enhanced monitoring possibilities in the P&L side.

Please advise the company accountant before using this settings.

Production Cost Information

Production Order			
Type	Standard	No.	492
Status	Released	Order Date	02.10.
Product No.	D00002	Start Date	02.10.
Planned Qty	2	Due Date	02.10.
Warehouse	01	...	
Components Summary			
<u>Costs</u>		<u>Quantities</u>	<u>Planned Times</u>
Actual Item Component Cost	120	Planned Quantity	2
Actual Resource component costs	50	Completed Quantity	...
Actual Additional Cost		Rejected Quantity	
Actual Product Cost	200		
Actual By-Product Cost		<u>Dates</u>	
Total Variance	30	Due Date	02.10.
		Actual Closing Date	02.10.
		Overdue	
Journal Remark	Production Order – D00002		
...			

On the *Summary* tab page you have the information that summarizes the *Production Order*. The tab displays the information on the released and closed *Production Orders*.

The accountant refers to the *Summary* tab when she wants to check cost information.

Actual Item Component Cost: Displays the total value of all item components issued to the *Production Order*.

Actual Resource Component Cost: Displays the total cost of all the resource components used in the *Production Order*.

Actual Additional Cost: Displays the total of the actual additional costs of the non-inventory items. This can be, for example, a service or labor cost. Note that this cost is not related to the *Additional Quantity* in the BOM.

Actual Product Cost: Displays the posted value of the finished product which is the best estimate calculated by the system at the time of posting. In our example the cost of the components (items and resources) do not sum to the posted parent cost.

Actual By-Product Cost: Displays the total cost of all By product.

Total Variance: Displays the value that is the result of the total actual components value subtracted from the total actual products value.

Variance report

➔ From a *Production Order* → *Summary* tab → *Total Variance* field link arrow

Variance Report						
Product = Wooden Door Planned quantity = 10 Actual quantity = 9						
#	Type	Description	Qty	Avg. Cost	Total	Variance
1	Item component	Wooden plate	-10	50	-500	-50
2	Item component	Handle	-20	10	-200	-20
3	Resource component	Lathe Machine	-20	20	-400	-40
4	Product	Wooden Door	9	110	990	0
5	Product revaluation	Wooden Door	0	0	110	110

Only 9 wooden doors were received into inventory

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9

The *Variance Report* displays the breakdown of costs and quantities related to a specific *Production Order* throughout its production process.

In addition, it also shows the total value added and deducted from Inventory during production processing.

Let us examine the scenario in the table displayed. In this case, there is a cost variance since the actual parent item quantity is less than the planned one. The planned components for the production of **10** doors were fully consumed (rows 1-3). However, only **9** doors were actually produced and received into inventory (row 4). When the production order was closed, a posting was added to adjust the total parent value for the consumption variance (row 5).

Key Points



Key points from this topic:

- When working with perpetual inventory several journal entries are posted during the production process.
- A transaction is added when adding an *Issue for Production*, a *Receipt from Production* and when closing the *Production Order*.
- If required, a P&L WIP accounts can also be involved in the production journal entries.
- There are several production scenarios that cause a variance in the final product cost. This variance is reflected in the production order's closing journal entry.
- In the *Summary* tab of the *Production Order*, there is a summary of the product and components costs.
- The *Variance Report* displays the breakdown of costs and quantities related to a specific *Production Order* throughout its production process

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